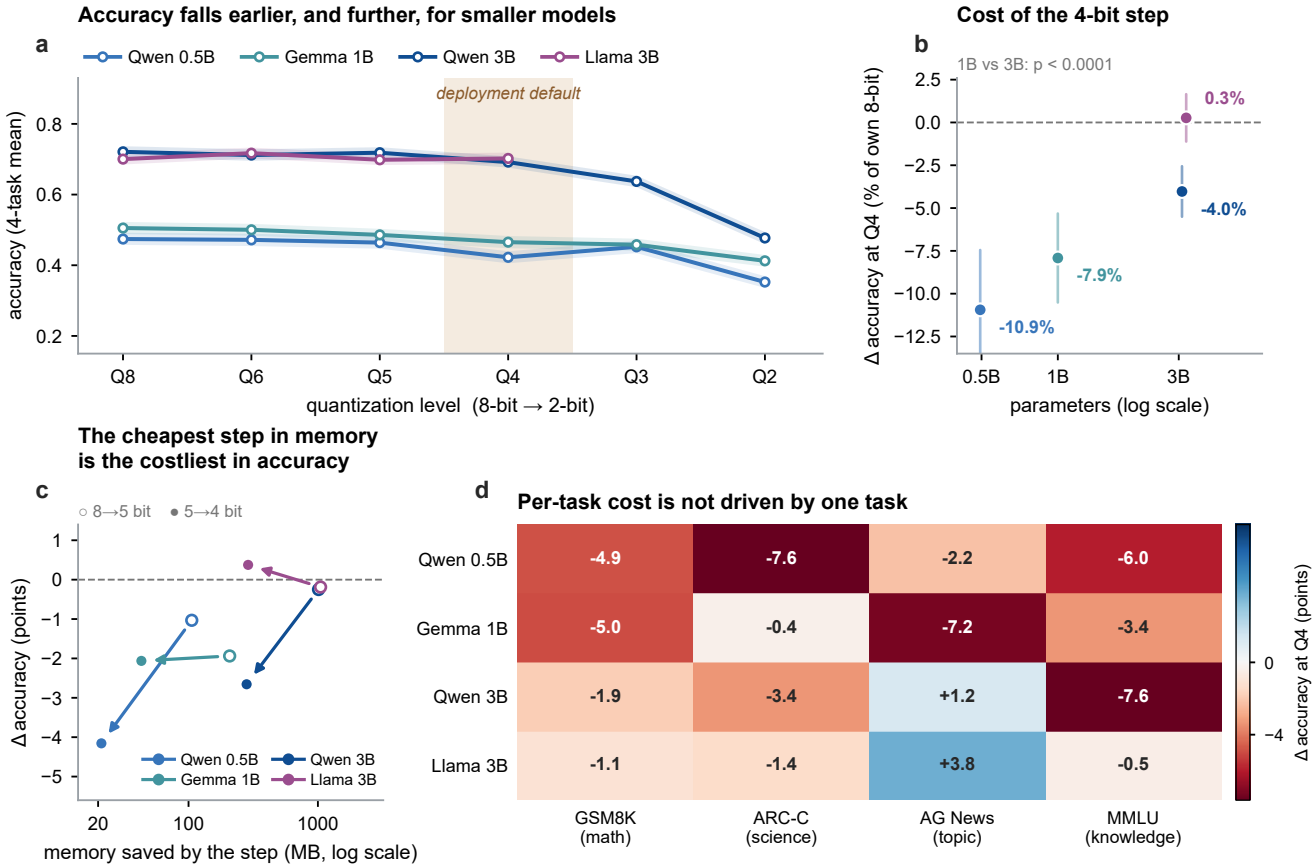

Accuracy falls earlier, and further, for smaller models
a
Qwen 0.5B
Gemma 1B
Qwen 3B
Llama 3B
deployment default
accuracy (4-task mean)
0.8
0.6
0.4
0.2
Q8
Q6
Q5
Q4
Q3
Q2
quantization level (8-bit → 2-bit)
Cost of the 4-bit step
b
1B vs 3B: p < 0.0001
Δ accuracy at Q4 (% of own 8-bit)
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
0.3%
-4.0%
-7.9%
-10.9%
0.5B
1B
3B
parameters (log scale)
The cheapest step in memory
is the costliest in accuracy
c
○ 8→5 bit
● 5→4 bit
Δ accuracy (points)
1
0
−1
−2
−3
−4
−5
Qwen 0.5B
Qwen 3B
Gemma 1B
Llama 3B
20
100
1000
memory saved by the step (MB, log scale)
d
Per-task cost is not driven by one task
Qwen 0.5B
Gemma 1B
Qwen 3B
Llama 3B
-4.9
-7.6
-2.2
-6.0
-5.0
-0.4
-7.2
-3.4
-1.9
-3.4
+1.2
-7.6
-1.1
-1.4
+3.8
-0.5
GSM8K
(math)
ARC-C
(science)
AG News
(topic)
MMLU
(knowledge)
Δ accuracy at Q4 (points)
0
−4